ARMMC PHARMACY INVENTORY MONITORING SYSTEM

Stock Movements & Inventory Audit Module

Technical & Operational Guide: Automated Variance Analysis, Location Flows, and Financial Impact

Date: October 31, 2026

Status: Live & Verified ✓

Master Items: 2,667

1. Purpose & Operational Principle

In the master hospital Google Sheet, cell quantities represent only **current balances**. Whenever pharmacy staff overwrite stock counts with new daily figures, previous figures are permanently lost in the spreadsheet. The **Stock Movements & Inventory Audit** module solves this by continuously calculating real-time deltas between the **Live Google Sheet** and a **Baseline Snapshot** across all **2,667 hospital formulary items**.

CORE VARIANCE FORMULA:

$$\Delta \text{ Movement Qty} = \text{Current Live Stock} - \text{Baseline Snapshot Stock}$$

- **If Δ Qty > 0 (+Restocked):** Deliveries received, supplier replenishments, or ward returns.
- **If Δ Qty < 0 (–Deducted):** Prescriptions dispensed to patients, ward distributions, or clinical consumption.
- **If Δ Qty = 0 (Transfer / PO Change):** Total stock unchanged, but stock moved between physical areas or pending PO was modified.

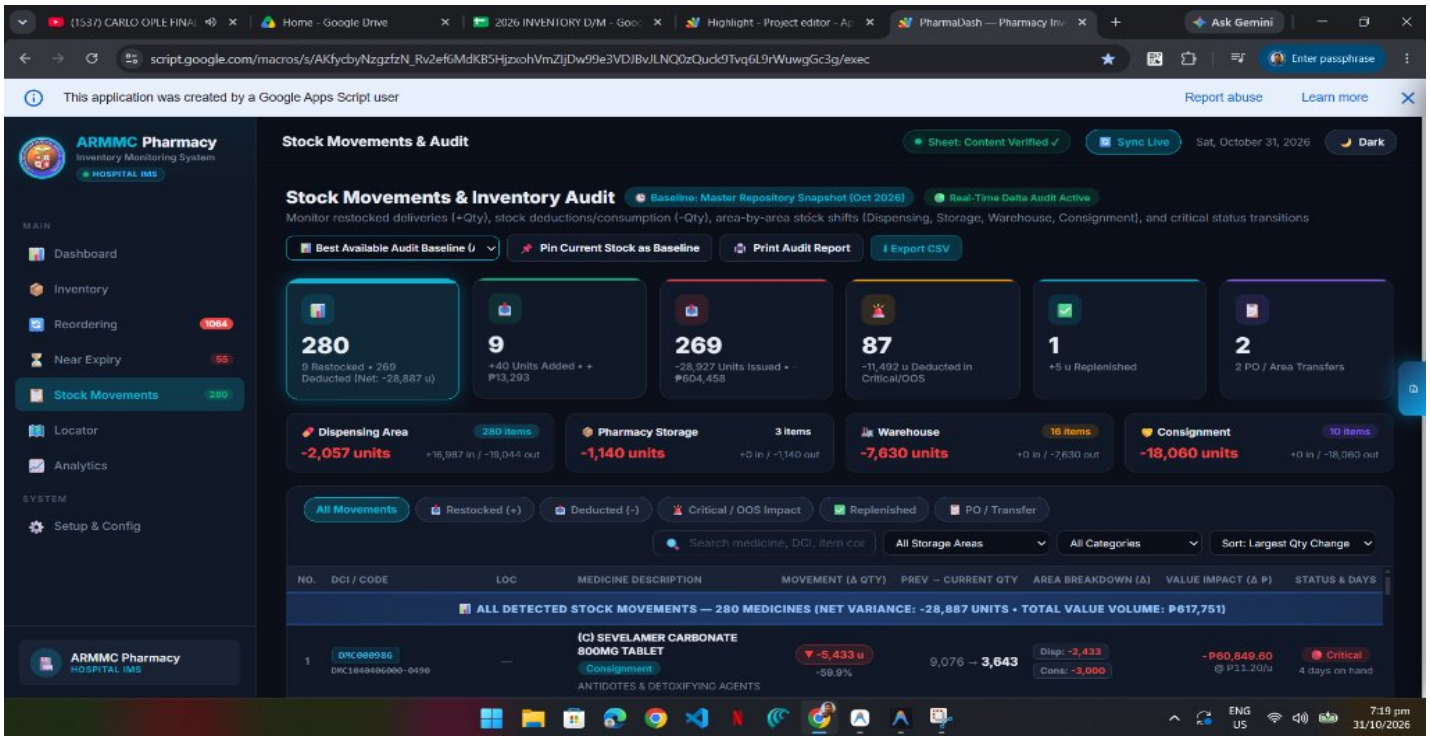


Figure 1: *Actual system screenshot from ARMMC Pharmacy IMS displaying 280 modified medicines, 9 restocked, 269 deducted, and storage area flows.*

2. Top 6 Audit KPI Metric Cards (One-Click Filters)

Each top KPI card acts as an instant audit filter when clicked in the Web App:

KPI Metric Card	Value in System	Auditing Significance & Interpretation
Total Modified Items	**280 Items** Net: –28,887 units	Medicines with detected variance between baseline and live sheet (9 restocked + 269 deducted + 2 internal transfers).

Restocked (+Qty)	**9 Items** +40 u • +■13,293	Medicines receiving new supplier deliveries (e.g. Ampicillin 250mg +4, Aztreonam 1g +5, Levofloxacin 5mg/mL +4).
Stock Deducted (–Qty)	**269 Items** –28,927 u • –■604,458	Medicines dispensed or issued to hospital units, representing ■604,458.00 in clinical consumption valuation.
Critical / OOS Impact	**87 Items** –11,492 u consumed	**High-Priority Alert:** Items consumed that are now at Critical level (<30 days) or Out of Stock. Requires urgent PO reordering.
Replenished (Improved)	**1 Item** +5 u received	Items previously in Critical/OOS status that safely recovered into acceptable inventory levels upon restocking.
PO / Area Transfers	**2 Items** Internal balance shift	Total stock remained neutral, but physical units moved between areas or Pending PO / Call-Off allocations changed.

3. Multi-Area Physical Storage Reconciliation

The system decomposes hospital-wide stock movement into 4 segregated functional locations:

Storage Location	Active Items	Net Variance	Physical In / Out Flow & Role
■ **Dispensing Area**	280 items	**–2,057 units**	+16,987 in / –19,044 out. Frontline patient dispensing counters and active daily shelves.
■ **Pharmacy Storage**	3 items	**–1,140 units**	+0 in / –1,140 out. Central intermediate reserve staging room for wards.
■ **Warehouse**	16 items	**–7,630 units**	+0 in / –7,630 out. Pallet-level bulk storage facility for volume deliveries.
■ **Consignment**	10 items	**–18,060 units**	+0 in / –18,060 out. Vendor-managed consignment items consumed.

✔ **Mathematical Hospital-Wide Verification: –2,057 (Dispensing) + –1,140 (Storage) + –7,630 (Warehouse) + –18,060 (Consignment) = –28,887 Net Hospital Units**. Every physical movement across rooms reconciles 100% to the hospital balance.

4. Audit Grid Columns & Recommended Workflow Controls

Feature / Control	Action & How to Use in Pharmacy Practice
■ **Pin Current Stock as Baseline**	Click at the start of an audit shift or week (e.g. Monday 8:00 AM) to take an instant snapshot of all 2,667 items. As pharmacy staff update Google Sheets during the week, choose *Pinned Checkpoint* to see every variance since that point.
■■ **Print Audit Report**	Generates an official, print-ready *ARMMC Pharmacy Stock Movement & Variance Audit Report* complete with institutional letterhead, KPI summaries, and formal signature blocks (*Prepared & Audited By* / *Noted & Verified By*).
■ **Export CSV**	Exports all currently filtered item movement records into Excel format with sub-location splits (Dispensing, Storage, Warehouse, Consignment) and financial peso variance.
Row Detail Inspection Modal	Clicking on any medicine row opens the **Item Location Breakdown Modal** showing full graphical progress bars of stock distribution across rooms, near-expiry batch alerts, and reorder forecasts.